Unit 19: Monotonic Functions and the First Derivative Test — Full Solutions

Warm-up

1. $f(x) = x^2 - 4x + 1$

$f'(x) = 2x - 4$. Critical point: $x = 2$.

Test $x < 2$ (e.g., $x = 0$): $f'(0) = -4 < 0 \to$ decreasing. Test $x > 2$ (e.g., $x = 3$): $f'(3) = 2 > 0 \to$ increasing.

Decreasing on $(-\infty, 2)$, increasing on $(2, \infty)$. Local minimum at $x = 2$ (sign goes $-$ to $+$).

2. $f(x) = -x^2 + 6x - 5$

$f'(x) = -2x + 6$. Critical point: $x = 3$.

Test $x < 3$: $f'(0) = 6 > 0 \to$ increasing. Test $x > 3$: $f'(4) = -2 < 0 \to$ decreasing.

Increasing on $(-\infty, 3)$, decreasing on $(3, \infty)$. Local maximum at $x = 3$.

3. $f(x) = x^3 - 3x$

$f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$. Critical points: $x = 1$, $x = -1$.

Test $x < -1$ (e.g., -2): $3(-3)(-1) = 9 > 0 \to$ increasing. Test $-1 < x < 1$ (e.g., 0): $3(-1)(1) = -3 < 0 \to$ decreasing. Test $x > 1$ (e.g., 2): $3(1)(3) = 9 > 0 \to$ increasing.

Increasing on $(-\infty, -1)$, decreasing on $(-1, 1)$, increasing on $(1, \infty)$. Local max at $x = -1$; local min at $x = 1$.

4. $f(x) = x^3 - 12x$

$f'(x) = 3x^2 - 12 = 3(x-2)(x+2)$. Critical points: $x = 2$, $x = -2$.

Test $x < -2$ (-3): $3(-5)(-1) = 15 > 0 \to$ increasing. Test $-2 < x < 2$ (0): $3(-2)(2) = -12 < 0 \to$ decreasing. Test $x > 2$ (3): $3(1)(5) = 15 > 0 \to$ increasing.

Increasing on $(-\infty, -2)$, decreasing on $(-2, 2)$, increasing on $(2, \infty)$. Local max at $x = -2$; local min at $x = 2$.

5. $f(x) = x^2 + 2x - 3$

$f'(x) = 2x + 2$. Critical point: $x = -1$.

Test $x < -1$: $f'(-2) = -2 < 0 \to$ decreasing. Test $x > -1$: $f'(0) = 2 > 0 \to$ increasing.

Decreasing on $(-\infty, -1)$, increasing on $(-1, \infty)$. Local minimum at $x = -1$.

6. $f(x) = 2x^3 - 6x$

$f'(x) = 6x^2 - 6 = 6(x-1)(x+1)$. Critical points: $x = 1$, $x = -1$.

Test $x < -1$ (-2): $6(-3)(-1) = 18 > 0 \to$ increasing. Test $-1 < x < 1$ (0): $6(-1)(1) = -6 < 0 \to$ decreasing. Test $x > 1$ (2): $6(1)(3) = 18 > 0 \to$ increasing.

Increasing on $(-\infty, -1)$, decreasing on $(-1, 1)$, increasing on $(1, \infty)$. Local max at $x = -1$; local min at $x = 1$.

Standard

7. $f(x) = x^3 - 3x^2 - 9x + 5$

$f'(x) = 3x^2 - 6x - 9 = 3(x-3)(x+1)$. Critical points: $x = 3$, $x = -1$.

Test $x < -1$ (-2): $3(-5)(-1) = 15 > 0 \to$ increasing. Test $-1 < x < 3$ (0): $3(-3)(1) = -9 < 0 \to$ decreasing. Test $x > 3$ (4): $3(1)(5) = 15 > 0 \to$ increasing.

Increasing on $(-\infty, -1)$, decreasing on $(-1, 3)$, increasing on $(3, \infty)$. Local max at $x = -1$; local min at $x = 3$.

8. $f(x) = x^4 - 4x^3$

$f'(x) = 4x^3 - 12x^2 = 4x^2(x-3)$. Critical points: $x = 0$, $x = 3$.

Test $x < 0$ (-1): $4(1)(-4) = -16 < 0 \to$ decreasing. Test $0 < x < 3$ (1): $4(1)(-2) = -8 < 0 \to$ decreasing (same sign as before — no change at $x = 0$). Test $x > 3$ (4): $4(16)(1) = 64 > 0 \to$ increasing.

Decreasing on $(-\infty, 0) \cup (0, 3)$ (i.e., decreasing throughout $(-\infty, 3)$), increasing on $(3, \infty)$. At $x = 0$: no sign change, so it's neither a local max nor min. Local min at $x = 3$.

9. $f(x) = 3x^4 - 4x^3 - 12x^2 + 1$

$f'(x) = 12x^3 - 12x^2 - 24x = 12x(x-2)(x+1)$. Critical points: $x = 0$, $x = 2$, $x = -1$.

Test $x < -1$ (-2): $x(x-2)(x+1) = (-2)(-4)(-1) = -8 < 0 \to$ decreasing.

Test $-1 < x < 0$ (-0.5): $(-0.5)(-2.5)(0.5) = 0.625 > 0 \to$ increasing.

Test $0 < x < 2$ (1): $(1)(-1)(2) = -2 < 0 \to$ decreasing.

Test $x > 2$ (3): $(3)(1)(4) = 12 > 0 \to$ increasing.

Decreasing on $(-\infty, -1)$, increasing on $(-1, 0)$, decreasing on $(0, 2)$, increasing on $(2, \infty)$. Local min at $x = -1$; local max at $x = 0$; local min at $x = 2$.

10. $f(x) = \dfrac{x}{x^2+1}$

Using the quotient rule: $f'(x) = \dfrac{1 \cdot (x^2+1) - x \cdot 2x}{(x^2+1)^2} = \dfrac{1-x^2}{(x^2+1)^2}$.

The denominator is always positive, so the sign of f' matches the sign of $1 - x^2 = (1-x)(1+x)$. Critical points: $x = 1$, $x = -1$.

Test $x < -1$ (-2): $1 - 4 = -3 < 0 \to$ decreasing. Test $-1 < x < 1$ (0): $1 - 0 = 1 > 0 \to$ increasing. Test $x > 1$ (2): $1 - 4 = -3 < 0 \to$ decreasing.

Decreasing on $(-\infty, -1)$, increasing on $(-1, 1)$, decreasing on $(1, \infty)$. Local min at $x = -1$; local max at $x = 1$.

11. $f(x) = (x-1)^2(x+2)$

Using the product rule with $u = (x-1)^2$ $(u' = 2(x-1))$ and $v = (x+2)$ $(v' = 1)$:

$$f'(x) = 2(x-1)(x+2) + (x-1)^2(1) = (x-1)[2(x+2) + (x-1)] = (x-1)(3x+3) = 3(x-1)(x+1)$$

Critical points: $x = 1$, $x = -1$.

Test $x < -1$ (-2): $3(-3)(-1) = 9 > 0 \rightarrow$ increasing. Test $-1 < x < 1$ (0): $3(-1)(1) = -3 < 0 \rightarrow$ decreasing. Test $x > 1$ (2): $3(1)(3) = 9 > 0 \rightarrow$ increasing.

Increasing on $(-\infty, -1)$, decreasing on $(-1, 1)$, increasing on $(1, \infty)$. Local max at $x = -1$; local min at $x = 1$.

12. $f(x) = x^5 - 5x$

$f'(x) = 5x^4 - 5 = 5(x^4 - 1) = 5(x^2 - 1)(x^2 + 1)$. Since $x^2 + 1$ is always positive, the sign is determined by $(x^2 - 1) = (x-1)(x+1)$. Critical points: $x = 1$, $x = -1$.

Test $x < -1$ (-2): $(-3)(-1) = 3 > 0 \rightarrow$ increasing. Test $-1 < x < 1$ (0): $(-1)(1) = -1 < 0 \rightarrow$ decreasing. Test $x > 1$ (2): $(1)(3) = 3 > 0 \rightarrow$ increasing.

Increasing on $(-\infty, -1)$, decreasing on $(-1, 1)$, increasing on $(1, \infty)$. Local max at $x = -1$; local min at $x = 1$.

Challenge

13. $f(x) = x^3 - \frac{3}{2}x^2 - 6x + 3$

$f'(x) = 3x^2 - 3x - 6 = 3(x^2 - x - 2) = 3(x-2)(x+1)$. Critical points: $x = 2$, $x = -1$.

Test $x < -1$ (-2): $3(-4)(-1) = 12 > 0 \rightarrow$ increasing. Test $-1 < x < 2$ (0): $3(-2)(1) = -6 < 0 \rightarrow$ decreasing. Test $x > 2$ (3): $3(1)(4) = 12 > 0 \rightarrow$ increasing.

Increasing on $(-\infty, -1)$, decreasing on $(-1, 2)$, increasing on $(2, \infty)$.

Local max at $x = -1$: $f(-1) = -1 - \frac{3}{2} - (-6) + 3 = -1 - 1.5 + 6 + 3 = 6.5 = \frac{13}{2}$.

Local min at $x = 2$: $f(2) = 8 - \frac{3}{2}(4) - 12 + 3 = 8 - 6 - 12 + 3 = -7$.

14. $f(x) = x^{2/3}(x - 5)$

First, expand: $f(x) = x^{5/3} - 5x^{2/3}$.

$$f'(x) = \frac{5}{3}x^{2/3} - \frac{10}{3}x^{-1/3}$$

Factor out $\frac{5}{3}x^{-1/3}$:

$$f'(x) = \frac{5}{3}x^{-1/3}(x - 2) = \frac{5(x-2)}{3x^{1/3}}$$

Critical points: $x = 2$ (where the numerator is 0) and $x = 0$ (where the denominator is 0, making f' undefined).

Test $x < 0$ (-1): numerator $= 5(-3) = -15$; denominator $= 3(-1)^{1/3} = 3(-1) = -3$. $f' = -15/-3 = 5 > 0 \rightarrow$ increasing.

Test $0 < x < 2$ (1): numerator $= 5(-1) = -5$; denominator $= 3(1) = 3$. $f' = -5/3 < 0 \to$ decreasing.

Test $x > 2$ (3): numerator $= 5(1) = 5$; denominator $= 3(3^{1/3}) > 0$. $f' > 0 \to$ increasing.

Increasing on $(-\infty, 0)$, decreasing on $(0, 2)$, increasing on $(2, \infty)$.

At $x = 0$ (a cusp, since f' is undefined there): sign changes $+$ to $-$, so $x = 0$ is a local maximum, with $f(0) = 0$.

At $x = 2$: sign changes $-$ to $+$, so $x = 2$ is a local minimum, with $f(2) = 2^{2/3}(2-5) = -3 \cdot 2^{2/3} = -3\sqrt[3]{4} \approx -4.76$.

15. $f(x) = \sin x + \cos x$ on $[0, 2\pi]$

$f'(x) = \cos x - \sin x$. Setting this to 0: $\tan x = 1$, giving $x = \dfrac{\pi}{4}$ and $x = \dfrac{5\pi}{4}$ within $[0, 2\pi]$.

Test $[0, \pi/4)$ (e.g., $x = 0$): $f'(0) = 1 - 0 = 1 > 0 \to$ increasing.

Test $(\pi/4, 5\pi/4)$ (e.g., $x = \pi$): $f'(\pi) = -1 - 0 = -1 < 0 \to$ decreasing.

Test $(5\pi/4, 2\pi]$ (e.g., $x = \frac{3\pi}{2}$): $f'\left(\frac{3\pi}{2}\right) = 0 - (-1) = 1 > 0 \to$ increasing.

Increasing on $\left[0, \frac{\pi}{4}\right)$, decreasing on $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$, increasing on $\left(\frac{5\pi}{4}, 2\pi\right]$.

Local max at $x = \dfrac{\pi}{4}$, where $f\left(\dfrac{\pi}{4}\right) = \sqrt{2}$. Local min at $x = \dfrac{5\pi}{4}$, where $f\left(\dfrac{5\pi}{4}\right) = -\sqrt{2}$.

16. $f(x) = x^4 - 8x^2 + 3$

$f'(x) = 4x^3 - 16x = 4x(x^2 - 4) = 4x(x-2)(x+2)$. Critical points: $x = 0$, $x = 2$, $x = -2$.

Test $x < -2$ (-3): $4(-3)(-5)(-1) = -60 < 0 \to$ decreasing.

Test $-2 < x < 0$ (-1): $4(-1)(-3)(1) = 12 > 0 \to$ increasing.

Test $0 < x < 2$ (1): $4(1)(-1)(3) = -12 < 0 \to$ decreasing.

Test $x > 2$ (3): $4(3)(1)(5) = 60 > 0 \to$ increasing.

Decreasing on $(-\infty, -2)$, increasing on $(-2, 0)$, decreasing on $(0, 2)$, increasing on $(2, \infty)$. Local min at $x = -2$; local max at $x = 0$; local min at $x = 2$.

17. $f(x) = \dfrac{x^2 - 4}{x}$

Simplify first: $f(x) = x - \dfrac{4}{x} = x - 4x^{-1}$.

$$f'(x) = 1 + 4x^{-2} = 1 + \frac{4}{x^2}$$

Since $\dfrac{4}{x^2}$ is always positive for $x \neq 0$, $f'(x)$ is always greater than 1, meaning it's always positive — there's no x where $f'(x) = 0$. The only place f' is undefined is $x = 0$, which isn't in the domain of f at all (it's excluded, not a true critical point to classify).

Answer: f is increasing on $(-\infty, 0)$ and increasing on $(0, \infty)$ (each piece separately, since $x = 0$ is a vertical asymptote that breaks the domain). There are no local maxima or minima anywhere.

Applied

18. $P(x) = -x^3 + 15x^2 - 48x + 10$, $x \geq 0$

$P'(x) = -3x^2 + 30x - 48 = -3(x^2 - 10x + 16) = -3(x-2)(x-8)$. Critical points: $x = 2$, $x = 8$.

Test $0 \leq x < 2$ (1): $-3(-1)(-7) = -21 < 0 \to$ decreasing.

Test $2 < x < 8$ (5): $-3(3)(-3) = 27 > 0 \to$ increasing.

Test $x > 8$ (9): $-3(7)(1) = -21 < 0 \to$ decreasing.

Decreasing on $[0,2)$, increasing on $(2,8)$, decreasing on $(8,\infty)$. Local minimum at $x = 2$; local maximum profit at $x = 8$ units.

19. $P(t) = -t^3 + 9t^2 + 120$, t in $[0,10]$

$P'(t) = -3t^2 + 18t = -3t(t-6)$. Critical points: $t = 0$, $t = 6$.

Test $0 < t < 6$ (3): $-3(3)(-3) = 27 > 0 \to$ increasing.

Test $6 < t < 10$ (8): $-3(8)(2) = -48 < 0 \to$ decreasing.

Increasing on $(0,6)$, decreasing on $(6,10)$. Local (and here, absolute on this interval) maximum population at $t = 6$ years.

20. $h(t) = -16t^2 + 64t + 5$

$h'(t) = -32t + 64$. Critical point: $t = 2$.

Test $t < 2$: $h'(0) = 64 > 0 \to$ increasing. Test $t > 2$: $h'(3) = -32 < 0 \to$ decreasing.

Increasing on $(-\infty,2)$ (physically, $[0,2)$), decreasing on $(2,\infty)$. Local maximum height occurs at $t = 2$ seconds — the peak of the ball's flight.

21. $T(t) = t^3 - 6t^2 + 9t + 20$, t in $[0,5]$

$T'(t) = 3t^2 - 12t + 9 = 3(t-1)(t-3)$. Critical points: $t = 1$, $t = 3$.

Test $0 \leq t < 1$ (0.5): $3(-0.5)(-2.5) = 3.75 > 0 \to$ increasing.

Test $1 < t < 3$ (2): $3(1)(-1) = -3 < 0 \to$ decreasing.

Test $3 < t \leq 5$ (4): $3(3)(1) = 9 > 0 \to$ increasing.

Increasing on $[0,1)$, decreasing on $(1,3)$, increasing on $(3,5]$.

Local max at $t = 1$: $T(1) = 1 - 6 + 9 + 20 = 24°F$.

Local min at $t = 3$: $T(3) = 27 - 54 + 27 + 20 = 20°F$.